

PROJECT TRICORDER

Mk I · Field Operations Manual · Rev 0.6.32

On-device multimodal field instrument

Inquiry · vision · imagine · speech in/out · translator · audio watch · flora/fauna · sentry · dictation · readings · overhead cards · Memory Alpha · MEDICAL · GNSS clock · moon · hands-free voice menu

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Supersedes Rev 0.6.28 (13 September 2026)

1. Purpose and scope

This is the operator and installer document for Project Tricorder Mk I through software 0.6.32. It supersedes Rev 0.6.28. Material from 0.6.4–0.6.28 remains in force unless a later section replaces it. The on-device Help sheet is a short briefing. The HTML briefing shipped in the APK (manual.html) matches this revision. This Word file is the full operations manual.

NAVIGATION WARNING. The artificial horizon, G-meter, GPS speed, barometric altitude proxy, satellite plot, star map, SOL orbits, moon-phase panel, HF/VHF/UHF forecasts, and GNSS satellite clock are NOT certified by the FAA, NASA, EASA, ICAO, or any civil or military authority. They MUST NEVER be used as primary or backup navigation, attitude, timing, or approach references for any aircraft, rotorcraft, UAV, spacecraft, or surface vessel. The satellite clock is a field novelty.

SENTRY / AUDIO WATCH WARNING. Weapon, gunfire, and conflict alerts are keyword hits on SmolVLM captions or YAMNet AudioSet labels. They are not a weapons detector, not a 911 substitute, and will false-positive on toys, pictures, fireworks, and media.

KNOWLEDGE / MEDICAL NOTE. MEMORY ALPHA and MEDICAL are field briefings drawn from Wikipedia / WikiMed catalogs over Earth link and cached locally. They are not a physician and not Memory Alpha the Star Trek wiki.

DICTATION NOTE. Dictation and the hands-free menu use the same on-device Android SpeechRecognizer as Inquiry. Command verbs should be spoken in isolation when possible.

2. What changed since Rev 0.6.28

Rev 0.6.28 closed MEDICAL extracts, pinned airframe cards, and city BACK. Builds 0.6.30–0.6.32 add a voice GNSS clock with timers, moon phase on the Sol cycle, intelligent place names on logs and the horizon, and a three-state ROBO / hands-free top-level voice menu.

Build	What landed
0.6.30	CLOCK intercept. Android GnssClock UTC face (analog/digital, 12/24). Labeled timers with alarm. Sol third panel is moon phase. Logs, sensed events, and horizon show city/state + Open Location Code (plus code). Optional what3words key.
0.6.31	Clock face no longer blocked the mic. After CLOCK the recognizer arms itself. Clock utterances are consumed by the clock, not Gemma. Sheet sits in the upper half so Inquiry remains reachable.
0.6.32	ROBO cycles dark (Kokoro) → green ROBO → yellow [H/F R]. Yellow listens for TRICORDER + function. Modes still leave with the face button.

3. Face and spoken overlays

I Inquiry · II Sensor · III Sentry (continuous default) · IV Audio · T / XLATE Translator · H Horizon · S Sat/Stars · O Sol (space weather and orbits → earth forecast → moon → off) · L E V P logs · IPL / ALL abort · STT kill speech.

ROBO is no longer a two-state lamp. Dark = Kokoro. First touch = green ROBO (Android TextToSpeech for all speech out). Second touch = yellow [H/F R] (robot voice plus hands-free listen). Third touch returns dark and stops the listen loop.

Spoken overlays that do not require hands-free: SETTINGS, BACK END / BACKEND / CAMERA, HELP, DIAGNOSTICS, COMPUTER, CLOCK, DICTATION, MEMORY ALPHA, MEDICAL, SUMMARIZE READINGS, and the TRICORDER menu in §4.

4. Hands-free top-level menu (0.6.32)

4.1 Intent

The face already has more functions than will fit as new switches. Hands-free is a spoken directory, not a second UI. The operator arms it from ROBO, says TRICORDER and a function, and leaves that function with the existing button. Interior layout does not change.

4.2 Arming

Start from dark ROBO. Touch once for green robot voice. Touch again for yellow [H/F R]. The CRT lists the verbs. The listen lamp comes up about 1.6 seconds after the prompt so the unit does not hear itself.

4.3 Verb list

Utterance	Action
TRICORDER SENSOR	Engage II Sensor.
TRICORDER AUDIO	Engage IV Audio Sense.
TRICORDER SENTRY	Engage III Sentry (continuous).
TRICORDER TRANSLATOR / TRANSLATE	Engage translator.
TRICORDER HORIZON	Toggle the artificial horizon.
TRICORDER SATELLITES	Cycle satellites / stars.
TRICORDER SOL	Advance the Sol panel one step.
TRICORDER CLOCK	Open the GNSS clock sheet.
TRICORDER DICTATION	Enter dictation directory.
TRICORDER SETTINGS	Overlay Settings sheet.
TRICORDER BACK END	Overlay Backend and Camera sheet.
TRICORDER DIAGNOSTICS	Spoken health briefing.
TRICORDER SUMMARIZE READINGS	Readings chart (optional category and timeframe).
TRICORDER HELP	Help sheet plus a short spoken list.

The same lines work from Inquiry with hands-free off. Recognizer aliases include trick order, tric order, try corder, and tree corder.

4.4 What yellow ignores

While [H/F R] is on, speech that is not a TRICORDER line is discarded so a conversation in the room does not fire Gemma. Audio Sense, translator, clock, and dictation own the microphone while they are engaged; the hands-free loop waits until those modes are left by hand. Then it listens again.

5. Satellite clock (0.6.30–0.6.31)

5.1 Why it exists

It is an on-device, in-a-pinch clock that does not depend on a cell tower or NTP once GNSS has a lock. Existing field-log timestamps were deliberately left on the ordinary millisecond wall clock.

5.2 How to use

Say CLOCK in Inquiry, or TRICORDER CLOCK. After the face appears, wait for listen, then speak a command. analog / digital · twelve hour / twenty four hour · timer coffee five minutes · cancel timer coffee · cancel all · explain · terminate.

5.3 What the number is

When the receiver reports a valid full bias, GPS time is TimeNanos minus FullBiasNanos minus BiasNanos. Leap seconds convert that to UTC. The GNSS Measurement API reports nanoseconds. On a Galaxy Fold 3 in open sky the chip uncertainty is typically tens to a few hundred nanoseconds. That is not a laboratory clock and must not be treated as one. The hardware epoch is about 1 Hz; the face interpolates to four paints per second using elapsed realtime. Until lock, device time is shown and the explainer says so.

5.4 Timers

Each timer has a spoken label and a duration. At expiry the unit sounds an alarm and speaks the label. Several timers may run together. They live only in this session.

6. Moon phase and places (0.6.30)

6.1 Sol cycle

First press: current planetary orbits around Sol, space weather, and HF / VHF / UHF (including a full UHF aurora line). Second press: local earth forecast — about six hours plus today and the next two days — with a scrollbar. Third press: moon phase — drawn icon, illumination, days remaining in this phase, the phase that just ended, and the next phase. Computed on-device from a synodic month of 29.53058867 days. Fourth press: off. Voice MOON or MOON PHASE walks Sol to the moon panel.

6.2 Intelligent location

Every new log line, sensed event, and the horizon place field carry: raw GPS when available; city and region from Android Geocoder when the geocoder has a name; an offline Open Location Code (plus code) as the free analog of what3words. A what3words API key may be stored in settings. Replay uses the same stored place string so a later review still names the site.

7. Knowledge banks and overhead card (retained 0.6.26–0.6.28)

MEMORY ALPHA: say memory alpha housefly or bare MEMORY ALPHA then the topic. Simple English Wikipedia extracts and a lead image, cached under app files/knowledge/articles. MEDICAL: WikiMed is a title catalog only; body text and photographs are English Wikipedia extracts because mdwiki REST is Cloudflare-blocked from the APK. Not a clinician.

Overhead flight card (Audio Sense + aircraft checkbox + Earth link): OpenSky box around last GPS, adsbdb type/owner/route, photograph from Planespotters / airport-data / JetPhotos. Card stays until DISMISS AIRFRAME. FROM / TO opens a city dossier; BACK TO AIRFRAME returns the same craft so the other city can be opened. Spoken altitude and speed follow the Settings metric/English toggle. Robot voice, not Kokoro.

8. Readings, dictation, watch (retained)

SUMMARIZE READINGS [category] [day|week|month|all]. Letter bars A B C with QTY and a color legend. Touch a bar to speak that category and open a filtered log timeline on the same sheet.

Dictation wake: dictation or TRICORDER DICTATION. NEW, EDIT nnnn, LOG nnnn (CONFIRMED), SUMMARIZE nnnn, MINDMAP keyword, SPEAK nnnn, KILL nnnn (CONFIRMED), TERMINATE. ZULU then SAVE or ABORT. Buffer: CURSOR MINUS / PLUS word, BACKSPACE, NEWLINE, CUT / CUT WORD, PASTE, HOME / END. PUNCTUATION PERIOD (also bare PERIOD / DOT).

Sentry captions every dwell. Sure weapon: klaxon + WEAPON DETECTED spoken three times on robot voice. Possible weapon: chime. Motion does not gate vision. Audio watch priority: gunfire, scream, siren, air/rail/ship, vehicles, named animal, humans, entertainment. IPL / ALL aborts in-flight work. STT slider kills speech only. Kokoro text longer than 80 characters is chunked.

9. Imagine, translator, horizon, sky

Imagine: step and size sliders (256 / 384 / 512) against SD Turbo on 127.0.0.1:8188. The documented field use is a pictogram where there is no library of pictures — imagine a cave entrance while the translator asks the local about caves.

Translator: ML Kit packs, SOURCE / TARGET wheels, AUTO swap after each line. XLATE lamp green idle, yellow speaking, red thinking.

Horizon shows heading, live G, GPS speed, barometric altitude proxy, and the place line. Sat/Stars toggles live satellite groups and an overhead star map. Uncertified. See §1.

10. Field log kinds

Kind	When written
sensor / sentry	Still + caption. Flora line when the pack is on device. Place string attached.
audio_*	YAMNet / BirdNET watch hits with place.
flight	Overhead contact + photo JPEG + route extra + place.
flora	iNaturalist pack hits on a Sensor still.
officer	Spoken LOG ENTRY or LOG nnnn from dictation.
video	Front-camera clip when VIDEO LOG is armed.
context	NOAA / civic / remote mesh events.

11. Architecture

Kotlin WebView APK. Four Termux loopback services stay up together: Gemma chat :8080, SmolVLM vision :8081, SD Turbo imagine :8188, Kokoro TTS :5000. Launch them with ~/bin/tricorder start (ssh on 8022). Do not stop 8188 to free a port.

On-device, no extra port: YAMNet, BirdNET, iNaturalist flora/fauna (TFLite), ML Kit translator, Android SpeechRecognizer, Android TTS robot voice, Wikipedia/WikiMed REST + local article cache,

OpenSky / adsbdb / Planespotters, NWS CAP, Nearby Connections mesh, GnssClock, Geocoder, Open Location Code encoder.

12. Install and first run

Uninstall the previous Mk I package before sideloading 0.6.32 (same applicationId). Allow Camera, Microphone, Location (including GNSS measurements), Nearby devices. Termux from F-Droid, not Play. On first Earth-link boot the APK fetches BirdNET, flora packs, and translator language packs if they are missing, and says initializing tricorder. Backend install remains the Termux guide in docs/TERMUX-BACKEND-INSTALL.md.

13. Troubleshooting additions for 0.6.30–0.6.32

Symptom	Check
Clock face only, no timers	Need 0.6.31+. After CLOCK wait for the listen lamp. Say timer coffee five minutes. Yellow H/F also accepts TRICORDER CLOCK then the same verbs.
Clock shows device time	No GNSS lock yet (indoors). Walk outside. Explain reports lock, satellite count, and uncertainty.
[H/F R] does not engage Sensor	Must say TRICORDER before SENSOR. Wait for yellow listen. Aliases: trick order sensor.
Hands-free dies in Audio	By design. Audio Sense owns the mic. Leave Audio with IV, yellow listen returns.
Moon missing on Sol	Need 0.6.30+. Cycle O four times: space, earth, moon, off.
Log has lat/lon but no city	Geocoder needs a network fix at least once. Plus code is always encoded offline.
MEDICAL titles only	Need 0.6.28+. Body is English Wikipedia, not mdwiki.
Airframe flashes away	Need 0.6.28+. Use DISMISS AIRFRAME. Do not swipe the app.

14. Competing products

Phone LLM apps still ship one or two modalities. PocketPal is the same Kokoro graph Mk I uses for speech out. Offline translators ship language packs without a field instrument. ADS-B viewers show aircraft without a sentry or a spoken watch. Planetarium apps show stars without Gemma or flora. None combine Gemma + SmolVLM + SD Turbo + Kokoro + YAMNet/BirdNET/flora + offline translation + sentry/audio watch + mesh + overhead photography + spoken knowledge banks + a GNSS clock and a spoken top-level menu on one offline Snapdragon face. That combination is the point of Mk I.

15. Revision history

Rev	Date	Note
0.6.14	9 Sep 2026	Audio soundscape watch.
0.6.17	9 Sep 2026	Dictation token buffer.
0.6.21	10 Sep 2026	Flight route speech, ROBO, DIAGNOSTICS.
0.6.24	12 Sep 2026	Readings letters, TTS mute, overlay Settings.
0.6.26	13 Sep 2026	Flight photos, city/airframe facts, knowledge banks.
0.6.27	13 Sep 2026	One-shot knowledge search; flight card not clobbered.

0.6.28	13 Sep 2026	MEDICAL extracts; pinned airframe; city BACK.
0.6.30	15 Sep 2026	GNSS clock, moon phase, plus-code places.
0.6.31	15 Sep 2026	Clock auto-listen and command routing.
0.6.32	15 Sep 2026	Hands-free TRICORDER menu; three-state ROBO.

End of Rev 0.6.32. Keep this file with the APK and the project source zip. Project Tricorder is not authorized by or affiliated with The Wand Company, Paramount SkyDance, Desilu, Gulf and Western, CBS, Roddenberry Entertainment, or Wah Chang, which are trademarks of their respective holders.